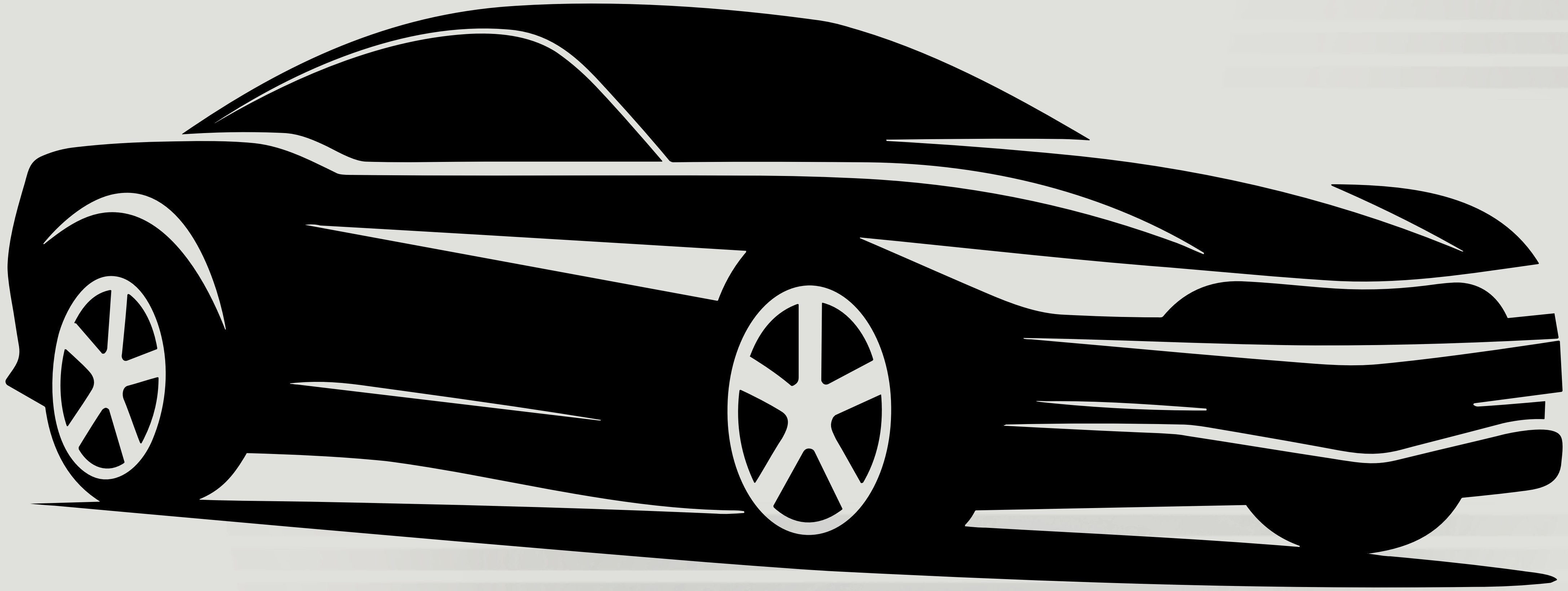
AI-POWERED PREDICTIVE MAINTENANCE FOR VEHICLES



MACHINE LEARNING PROJECT

THE PROBLEM

- Unexpected vehicle failures can cause:
    - High repair costs
    - Vehicle downtime
    - Safety risks
    - Disrupted operations
- Traditional maintenance approaches:
    - Reactive: Repair after failure
    - Preventive: Repair according to a fixed schedule

OUR SOLUTION

Vehicle Data → Machine Learning → Failure Prediction → Maintenance Action

The system analyses:

- Engine temperature
- Vibration levels
- Tyre pressure
- Battery status
- RPM
- Usage hours
- Vehicle age
- Maintenance information

BUSINESS VALUE

For Vehicle Owners

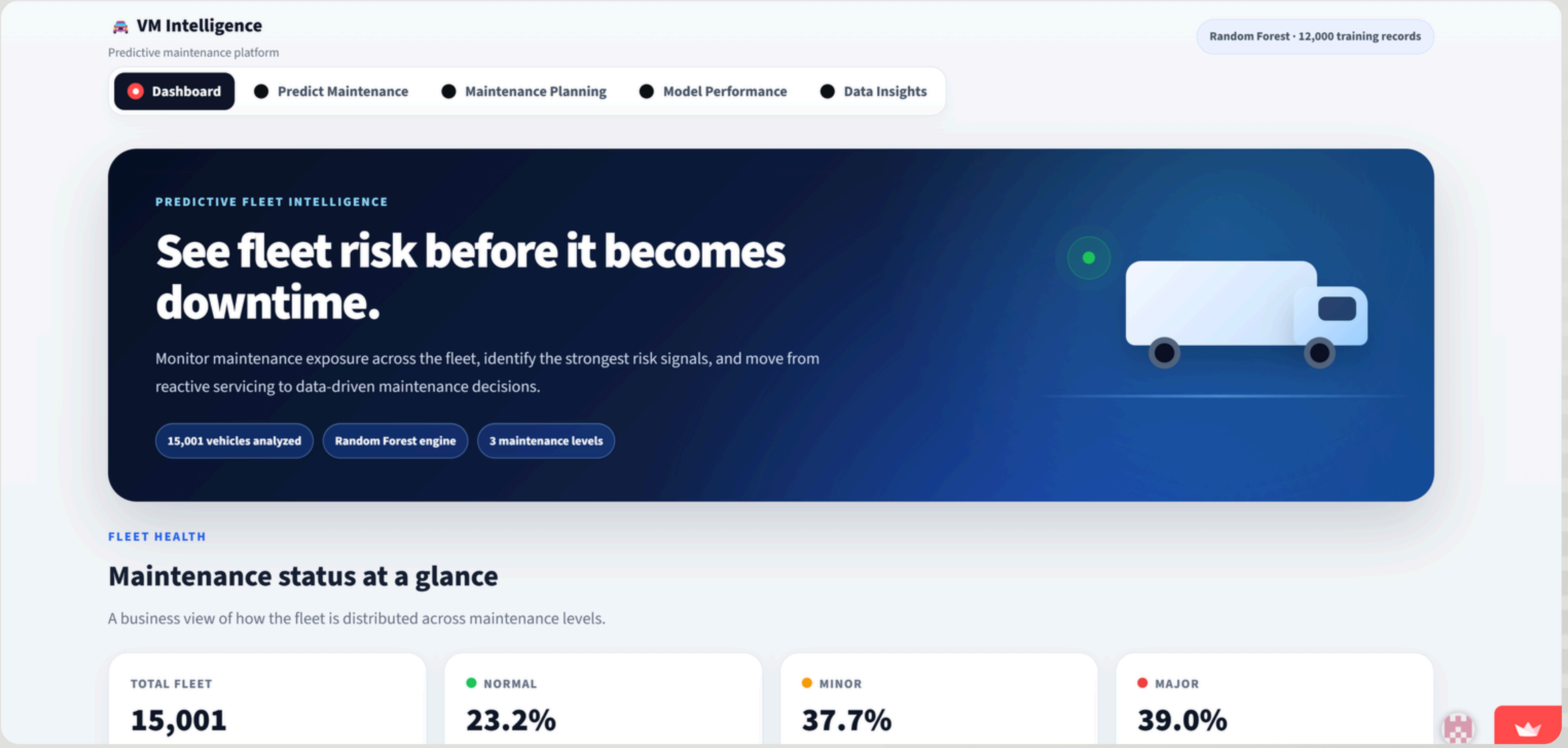
- Reduce unexpected repair costs
- Improve vehicle reliability
- Avoid sudden breakdowns

For Manufacturers

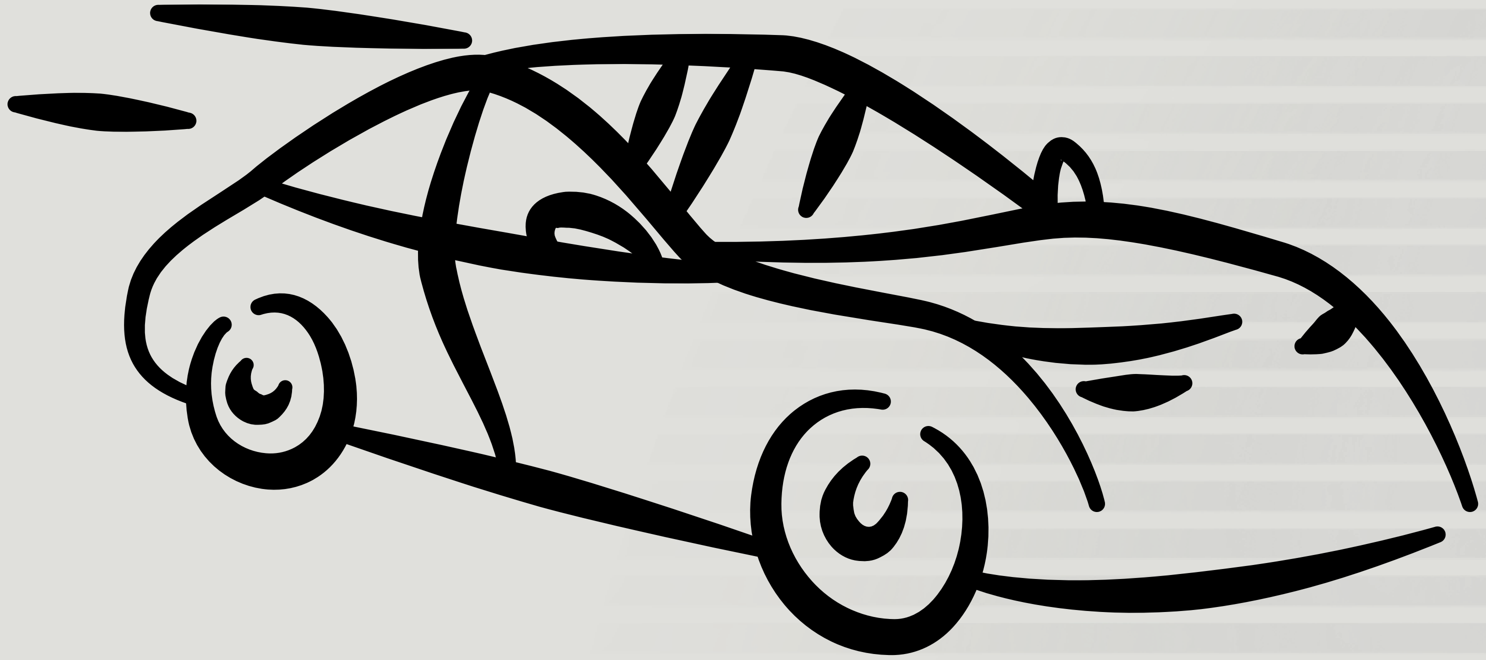
- Identify failure patterns
- Improve vehicle reliability
- Support data-driven decisions

Our Value Proposition:

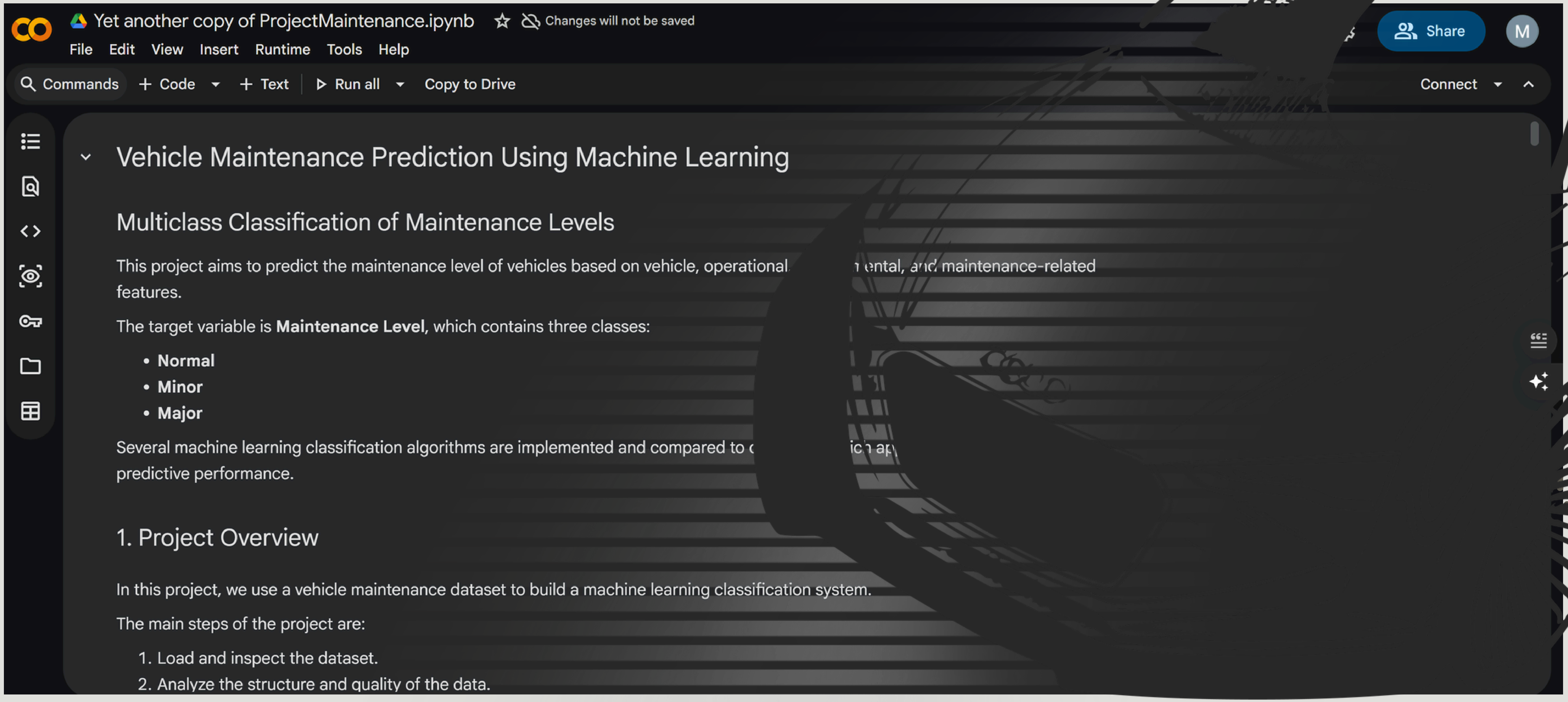
Predict failures earlier. Reduce downtime. Lower maintenance costs.



USER INTERFACE(UI)



THE MODEL



Vehicle Maintenance Prediction Using Machine Learning

Multiclass Classification of Maintenance Levels

This project aims to predict the maintenance level of vehicles based on vehicle, operational ental, and maintenance-related features.

The target variable is **Maintenance Level**, which contains three classes:

- **Normal**
- **Minor**
- **Major**

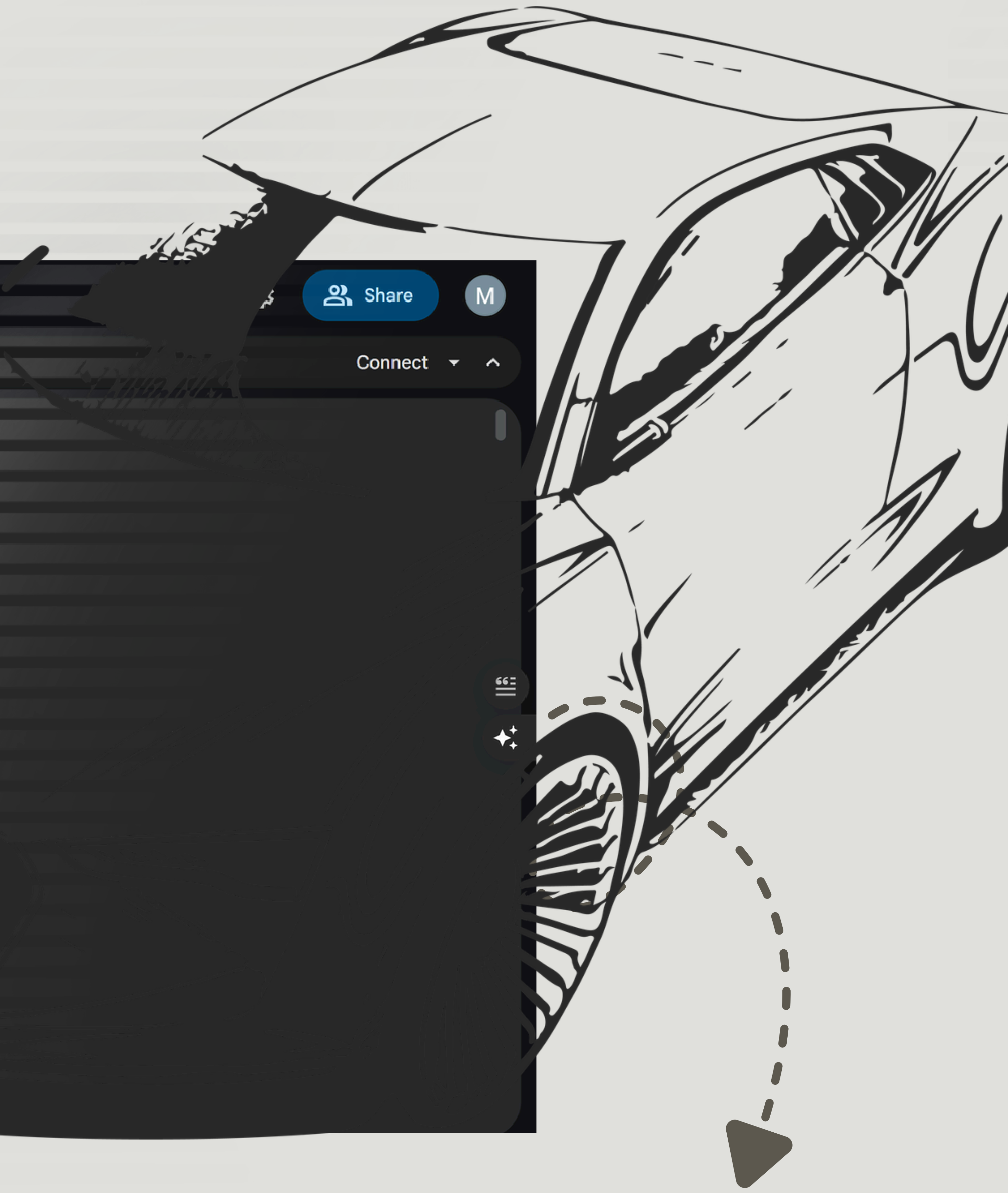
Several machine learning classification algorithms are implemented and compared to ich ap predictive performance.

1. Project Overview

In this project, we use a vehicle maintenance dataset to build a machine learning classification system.

The main steps of the project are:

1. Load and inspect the dataset.
2. Analyze the structure and quality of the data.



[Take a Look]

THANKS